WORKED SOLUTION

STEP I 1996, Question 4

Pure • Integration by Completing Square

STEP I 1996, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q4

The key idea

Whenever a quadratic sits in a denominator, complete the square: $x^2 + 2ax + 1 = (x + a)^2 + (1 - a^2)$. The sign of $1 - a^2$ decides everything. If it is positive we have a "sum of squares" and an inverse tangent; if it is negative we have a difference of squares, which splits into partial fractions and gives logarithms. The real work is then tidying the answers into the given forms.

Solution

Throughout, write I for the integral over $[0, 1]$. Completing the square,

$$x^2 + 2ax + 1 = (x + a)^2 + 1 - a^2.$$

The case $|a| < 1$

Put $b = \sqrt{1 - a^2} > 0$. Using $\int \frac{\mathrm{d}u}{u^2 + b^2} = \frac{1}{b} \tan^{-1} \frac{u}{b}$ with $u = x + a$,

$$I = \frac{1}{b}\left[\tan^{-1} \frac{x + a}{b}\right]_0^1 = \frac{1}{b}(\alpha - \beta), \qquad \text{where } \alpha = \tan^{-1} \frac{1 + a}{b}, \quad \beta = \tan^{-1} \frac{a}{b}.$$

(The denominator is never zero on $[0, 1]$, since it is at least $b^2 > 0$, so there is no trouble with the integral.) We need a single inverse tangent, so we take the tangent of $\alpha - \beta$:

$$\tan(\alpha - \beta) = \frac{\dfrac{1 + a}{b} - \dfrac{a}{b}}{1 + \dfrac{(1 + a)a}{b^2}} = \frac{b}{b^2 + a + a^2} = \frac{b}{1 + a},$$

using $b^2 + a^2 = 1$. Finally

$$\frac{b}{1 + a} = \frac{\sqrt{(1 - a)(1 + a)}}{1 + a} = \sqrt{\frac{1 - a}{1 + a}},$$

which is valid because $1 + a > 0$.

Ask yourself Does $\tan(\alpha-\beta)=t$ really give $\alpha-\beta=\tan^{-1}t$?

Only if $\alpha-\beta$ lies in $(-\frac{\pi}{2},\frac{\pi}{2})$, the range of $\tan^{-1}$, so we must check this. Since $(1+a)/b>0$ we have $0<\alpha<\frac{\pi}{2}$, and $-\frac{\pi}{2}<\beta<\frac{\pi}{2}$. Also $\alpha>\beta$, because $\tan^{-1}$ is increasing and $(1+a)/b>a/b$. So $0<\alpha-\beta<\pi$. On that interval tan is positive only on $(0,\frac{\pi}{2})$, and we found $\tan(\alpha-\beta)>0$; hence $0<\alpha-\beta<\frac{\pi}{2}$ and

$$I=\frac{1}{\sqrt{1-a^2}}\tan^{-1}\sqrt{\frac{1-a}{1+a}}\qquad(|a|<1),$$

as required.

(A slicker view: with $a=\cos\varphi$, $0<\varphi<\pi$, we get $\alpha=\frac{\pi}{2}-\frac{\varphi}{2}$ and $\beta=\frac{\pi}{2}-\varphi$, so $\alpha-\beta=\frac{\varphi}{2}$, and indeed $\sqrt{(1-\cos\varphi)/(1+\cos\varphi)}=\tan\frac{\varphi}{2}$.)

The case $|a|>1$

Put $c=\sqrt{a^2-1}>0$. Now the denominator is a difference of two squares:

$$x^2+2ax+1=(x+a)^2-c^2=(x+a-c)(x+a+c),$$

and partial fractions give

$$\frac{1}{(x+a-c)(x+a+c)}=\frac{1}{2c}\left(\frac{1}{x+a-c}-\frac{1}{x+a+c}\right).$$

The roots of the quadratic are $-a\pm c$. Their product is 1 and their sum is $-2a$.

Suppose first that $a>1$. Then $a>c$ (as $a^2>a^2-1$), so both $x+a-c$ and $x+a+c$ are positive for $x\geq 0$. Hence

$$I=\frac{1}{2c}\left[\ln\frac{x+a-c}{x+a+c}\right]_0^1=\frac{1}{2c}\left(\ln\frac{1+a-c}{1+a+c}-\ln\frac{a-c}{a+c}\right).$$

Both fractions simplify, using $(a-c)(a+c)=a^2-c^2=1$:

- $(1+a-c)(a+c)=(a+c)+(a-c)(a+c)=1+a+c$, so $\dfrac{1+a-c}{1+a+c}=\dfrac{1}{a+c}$;
- $a-c=\dfrac{1}{a+c}$, so $\dfrac{a-c}{a+c}=\dfrac{1}{(a+c)^2}$.

Therefore

$$I=\frac{1}{2c}\big(-\ln(a+c)+2\ln(a+c)\big)=\frac{1}{2\sqrt{a^2-1}}\ln\big(a+\sqrt{a^2-1}\big),$$

which is the stated result, since $a+\sqrt{a^2-1}>0$ here and so equals its modulus.

What about $a<-1$? Then the roots $-a\pm c=|a|\pm c$ are both positive with product 1, so the smaller one, $|a|-\sqrt{a^2-1}$, lies strictly between 0 and 1. The integrand has a simple pole inside $[0,1]$, behaving like a constant times $1/(x-r)$ near the root r. So

the integral does not converge, and strictly the formula holds only for $a > 1$. (The same partial-fraction calculation with $\ln|\cdot|$ shows that the formula gives the *Cauchy principal value*, the limit of the integral with a small symmetric gap around r removed. That explains the modulus signs in the question.)

Checks

- $a = 0$: the integral is $\int_0^1 \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{4}$, and the formula gives $\tan^{-1} 1 = \frac{\pi}{4}$.
- $a = 2$: Simpson's rule gives 0.380173, matching $\frac{1}{2\sqrt{3}} \ln(2+\sqrt{3})$.
- As $a \to 1$ both branches tend to $\frac{1}{2} = \int_0^1 (1+x)^{-2}\,\mathrm{d}x$. For the second, $\ln(a+c) \approx c$ when c is small; for the first, $\tan^{-1} t \approx t$ and $\sqrt{(1-a)/(1+a)}/\sqrt{1-a^2} = 1/(1+a)$.

Common slips

- Writing $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \frac{A-B}{1+AB}$ without checking that the difference lies in $(-\frac{\pi}{2}, \frac{\pi}{2})$.
- Forgetting that for $a < -1$ the denominator vanishes inside $[0,1]$.

Beyond STEP

The two answers are really one function of a. Writing $a = \cos\varphi$ in the first and $a = \cosh t$ in the second gives $\varphi/(2\sin\varphi)$ and $t/(2\sinh t)$, and with complex numbers ($\cosh t = \cos(\mathrm{i}t)$) one is the analytic continuation of the other. The divergence for $a < -1$ and the rescue by principal values are a first taste of how complex analysis handles poles on the path of integration. Needham's *Visual Complex Analysis* is a beautiful place to see this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.